# The Complete Treatise on Modern Psychology

A Game-Theoretic, Dynamical-Systems, and Machine-Learning Framework for Classification, Equilibrium, and Systemic Inversion

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**Abstract**

We develop a unified mathematical theory of psychological classification under strategic behaviour. Beginning with a two-type binary classifier (Visionary vs. Psychopath), we derive the mixed-strategy Nash equilibrium and its closed-form patient ratio $(1+b) : (1+a)$, where $a, b > 0$ are asymmetric misclassification costs. We then extend the model to four types—Visionary (V), Psychopath (P), Psychologist ($\Psi$), and Enigma ($\mathcal{E}$)—embedded in a Markov population process, and impose the closure constraint $\Psi = V + P + \mathcal{E}$ that makes the system self-referential. Optimal misclassification cost $\varepsilon^*$ for the Enigma type is derived as the geometric mean of a mimicry floor and a collapse ceiling. We characterise five qualitative regimes—stable, capture, cycling, epistemic collapse, and monoculture—and prove that the stability condition $\sigma < \sigma^*$ is both necessary and sufficient for the psychologist population to persist. Algorithms for simulation and equilibrium computation are provided alongside TikZ phase diagrams, and the treatise closes with machine-learning interpretations connecting the framework to mixture-of-experts models, adversarial classification, and Bayesian detection theory.

The treatise ends with "The End"

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# 1 Introduction

Classification is among the oldest problems in applied psychology. A practitioner observing a patient must assign a diagnostic label despite incomplete information, strategic behaviour on the patient's part, and the ever-present possibility that the classifier's own cognition is being shaped by the very population under study.

This treatise formalises that predicament. We model the interaction between a population of patients—drawn from types Visionary (V), Psychopath (P), and Enigma ($\mathcal{E}$)—and a population of classifying Psychologists ($\Psi$). All agents are rational, forward-looking, and capable of strategic signalling. The system is *closed*: the psychologist headcount equals the total patient headcount at every point in time, making the dynamics self-referential.

The contributions of this treatise are:

**C1.** A closed-form asymmetric Nash equilibrium for the binary classifier (Section 2).

**C2.** A four-type Markov population model with an explicit steady-state ratio and conservation law (Section 3).

**C3.** A derivation of the optimal Enigma misclassification cost $\varepsilon^*$ as the geometric mean of the mimicry floor $\varepsilon_1$ and the binding ceiling $\min(\varepsilon_2, \varepsilon_3)$ (Section 5).

**C4.** A complete regime map with necessary and sufficient stability conditions (Section 7).

**C5.** Machine-learning interpretations connecting to adversarial learning, mixture-of-experts, and Bayesian detection theory (Section 10).

# 2 The Binary Classification Game

## 2.1 Setup

Let $\mathcal{A} = \{V, P\}$ be the type space, with a continuum population of measure one. A Psychologist observes a patient and assigns a label $\hat{t} \in \mathcal{A}$. The patient simultaneously chooses a signal $s \in \mathcal{A}$. Both players choose mixed strategies.

**Definition 2.1** (Asymmetric payoff matrix)**.** Let $a > 0$ denote the cost of misclassifying a Psychopath as a Visionary (*danger cost*) and $b > 0$ denote the cost of misclassifying a Visionary as a Psychopath (*stigma cost*). The payoff matrix is:

$$\Pi = \begin{pmatrix} 1 & -b \\ -a & 1 \end{pmatrix}, \tag{1}$$

where rows index the patient's true type $(V, P)$ and columns index the psychologist's label $(V, P)$.

## 2.2 Nash Equilibrium

**Theorem 2.2** (Asymmetric binary equilibrium)**.** *In the unique mixed-strategy Nash equilibrium:*

$$p^* = \frac{1+b}{a+b+2}, \tag{2}$$

$$q^* = \frac{1+a}{a+b+2}, \tag{3}$$

*where $p^*$ is the probability the psychologist labels $V$ and $q^*$ is the probability the patient signals $V$.*

*Proof.* The psychologist is indifferent between labels when the patient mixes $q^*$:

$$q^*(1) + (1-q^*)(-a) = q^*(-b) + (1-q^*)(1). \tag{4}$$

Expanding: $q^* + aq^* - a = -bq^* + 1 - q^*$, so $q^*(a+b+2) = 1+a$, giving (3). Symmetrically, the patient is indifferent when $p^*(1) + (1-p^*)(-b) = p^*(-a) + (1-p^*)(1)$, yielding $p^*(a+b+2) = 1+b$, giving (2). Uniqueness follows because $\Pi$ has no pure-strategy equilibrium when $a, b > 0$ and the indifference conditions are linear with a unique solution. □

**Corollary 2.3** (Population ratio)**.** *For a closed population of $N$ patients, the expected equilibrium ratio* Visionary : Psychopath *is*

$$\boxed{V : P = (1+b) : (1+a).} \tag{5}$$

*Proof.* $V = Np^*$ and $P = N(1-p^*)$, so $V : P = p^* : (1-p^*) = (1+b) : (1+a)$. □

*Remark* 2.4. When $a = b$ (symmetric costs) the ratio collapses to 1:1, recovering the symmetric result. When $b > a$ (stigma more costly), the psychologist over-labels patients as V, paradoxically allowing more psychopaths to escape detection.

### 2.3 Comparative Statics

Table 1 records equilibrium outcomes for representative parameter values.

Table 1: Equilibrium ratios for $N = 100$ under selected cost parameters.

| $a$ | $b$ | $p^*$ | $q^*$ | $V$ | $P$ |
|---|---|---|---|---|---|
| 1 | 1 | 0.500 | 0.500 | 50 | 50 |
| 1 | 3 | 0.667 | 0.400 | 67 | 33 |
| 3 | 1 | 0.333 | 0.600 | 33 | 67 |
| 1 | 5 | 0.750 | 0.333 | 75 | 25 |
| 5 | 1 | 0.250 | 0.667 | 25 | 75 |
| 2 | 4 | 0.625 | 0.375 | 63 | 38 |

## 3 The Four-Type Extended Model

### 3.1 Type Space and Payoff Matrix

We extend $\mathcal{A} = \{V, P, \Psi, \mathcal{E}\}$. The Enigma type $\mathcal{E}$ combines visionary cognition with psychopathic strategy and can mimic any other type at will. The $4 \times 4$ payoff matrix is:

$$\Pi_4 = \begin{array}{c} \\ V \\ P \\ \Psi \\ \mathcal{E} \end{array} \begin{array}{c} \begin{array}{cccc} \hat{V} & \hat{P} & \hat{\Psi} & \hat{\mathcal{E}} \end{array} \\ \begin{pmatrix} 1 & -b & -b & -b \\ -a & 1 & -a & -a \\ -\gamma & -\delta & 1 & -\gamma \\ -\varepsilon & -\varepsilon & -\varepsilon & 1 \end{pmatrix} \end{array} \tag{6}$$

where $\gamma, \delta > 0$ are psychologist misclassification costs and $\varepsilon > 0$ is the symmetric Enigma cost, reflecting that no single label captures the Enigma.

### 3.2 Markov Population Dynamics

Let $\mathbf{x}(t) = (v, p, \psi, e)^\mathsf{T} \in \Delta^3$ be the population share vector at time $t$. The discrete-time dynamics are $\mathbf{x}(t+1) = M\,\mathbf{x}(t)$ with transition matrix:

$$M = \begin{pmatrix} 1-\lambda_v & 0 & \alpha & \mu_v \\ 0 & 1-\lambda_p & \beta & \mu_p \\ \lambda_v & \lambda_p & 1-\alpha-\beta-\phi & \mu_\psi \\ 0 & 0 & \phi & 1-\mu_v-\mu_p-\mu_\psi \end{pmatrix}, \tag{7}$$

where $\lambda_v, \lambda_p$ are patient-to-psychologist rates, $\alpha, \beta, \phi$ are psychologist outflow rates, and $\mu_v, \mu_p, \mu_\psi$ are Enigma reversion rates.

**Proposition 3.1** (Steady-state existence)**.** *For $M$ irreducible and aperiodic, a unique steady state $\mathbf{x}^*$ satisfying $M\mathbf{x}^* = \mathbf{x}^*$ exists and is given (up to normalisation) by:*

$$\begin{aligned} v^* &\propto \frac{\alpha\mu_\psi + \mu_v\phi}{\lambda_v}, \qquad p^* \propto \frac{\beta\mu_\psi + \mu_p\phi}{\lambda_p}, \\ \psi^* &\propto \mu_\psi, \qquad e^* \propto \phi. \end{aligned} \tag{8}$$

*Proof.* Write $M\mathbf{x} = \mathbf{x}$ component-wise:

$$\begin{aligned} \lambda_v v^* &= \alpha\psi^* + \mu_v e^*, \\ \lambda_p p^* &= \beta\psi^* + \mu_p e^*, \\ \phi\psi^* &= \mu_\psi e^*. \end{aligned}$$

From the third equation $e^* = (\phi/\mu_\psi)\psi^*$. Substituting into the first two gives $v^*$ and $p^*$ as stated. Normalising so that $v^* + p^* + \psi^* + e^* = 1$ completes the proof. □

The general steady-state ratio is therefore:

$$\boxed{V : P : \Psi : \mathcal{E} \;=\; \frac{\alpha\mu_\psi + \mu_v\phi}{\lambda_v} : \frac{\beta\mu_\psi + \mu_p\phi}{\lambda_p} : \mu_\psi : \phi.} \tag{9}$$

# 4 The Closed System: $\Psi = V + P + \mathcal{E}$

## 4.1 The Closure Constraint

Imposing $\psi^* = v^* + p^* + e^*$ on the steady-state ratios from (8), one obtains the *Conservation Law*:

$$\boxed{\mathcal{C} \;\equiv\; \frac{\alpha\mu_\psi + \mu_v\phi}{\lambda_v\mu_\psi} + \frac{\beta\mu_\psi + \mu_p\phi}{\lambda_p\mu_\psi} + \frac{\phi}{\mu_\psi} = 1.} \tag{10}$$

**Theorem 4.1** (Closure implies half-half split)**.** *Under the closure constraint, the population of $N$ agents splits exactly as $\psi = N/2$ psychologists and $v + p + e = N/2$ patients at every steady state.*

*Proof.* Total population $N = \psi + (v+p+e)$. Closure says $\psi = v+p+e$, hence $N = 2\psi$, i.e. $\psi = N/2$. □

## 4.2 Endogenous Transition Rates

The closure constraint forces the outflow rates $(\alpha, \beta, \phi)$ to be determined endogenously by equilibrium indifference conditions. Setting the psychologist's expected payoff from transitioning to each patient type equal to the payoff from remaining a psychologist:

$$\alpha = \frac{1+b}{2(a+b+2)}, \tag{11}$$

$$\beta = \frac{1+a}{2(a+b+2)}, \tag{12}$$

$$\phi = \frac{\varepsilon^*}{2(a+b+\varepsilon^*+2)}. \tag{13}$$

The *psychologist persistence condition* requires $\alpha + \beta + \phi < 1$.

## 4.3 The Master Equilibrium Equation

Substituting (11)–(13) into (10) and simplifying yields the single scalar equation governing all parameters:

$$\frac{(1+b)\lambda_p + (1+a)\lambda_v}{\lambda_v\lambda_p} + \frac{\varepsilon^*}{a+b+\varepsilon^*+2} = 2. \tag{14}$$

The left-hand side equals 2 precisely when the weighted misclassification burden equals twice the psychologist productivity — the system's self-sustaining condition.

# 5 Optimal Enigma Misclassification Cost

## 5.1 Three Stability Thresholds

Let $p_4$ denote the probability the psychologist labels a patient as $\mathcal{E}$ in the mixed-strategy equilibrium.

**Definition 5.1** (Stability thresholds)**.**

$$\varepsilon_1 = \frac{b-a}{1+a} \qquad \text{(mimicry floor)}$$

$$\varepsilon_2 = \frac{p_4 - \psi^* + \delta(1-\psi^*)}{1-p_4} \qquad \text{(transition ceiling)}$$

$$\varepsilon_3 = \frac{e^*(a+b+2) - (1-e^*)}{(1-e^*)(a+b+2) - 3} \qquad \text{(collapse ceiling)}$$

**Theorem 5.2** (Stability band)**.** *The classification system is stable if and only if*

$$\varepsilon_1 < \varepsilon < \min(\varepsilon_2, \varepsilon_3). \tag{15}$$

*Proof.* (*Necessity of* $\varepsilon > \varepsilon_1$.) The Enigma prefers revealing its true type over mimicking type $j$ iff $p_4 \geq p_j$ for all $j \neq 4$. This condition reduces to $\varepsilon > (b-a)/(1+a) = \varepsilon_1$ after substituting the equilibrium label probabilities from Theorem 2.2.

(*Necessity of* $\varepsilon < \varepsilon_2$.) The psychologist does not transition to $\mathcal{E}$ iff $U(\text{classify}) \geq U(\text{become } \mathcal{E})$. Solving for $\varepsilon$ yields $\varepsilon < \varepsilon_2$.

(*Necessity of* $\varepsilon < \varepsilon_3$.) The psychologist does not default to calling everyone $\mathcal{E}$ iff expected payoff of mixing over all labels exceeds $U(\text{always call } \mathcal{E})$. Solving yields $\varepsilon < \varepsilon_3$.

(*Sufficiency.*) When all three inequalities hold, each agent's strategy is a best response to the others and the Markov chain has a unique interior fixed point by Brouwer's theorem applied to the compact convex set $\Delta^3$. □

## 5.2 The Optimal Point

**Theorem 5.3** (Optimal $\varepsilon^*$)**.** *Within the stability band, the payoff-optimal misclassification cost is*

$$\boxed{\varepsilon^* = \sqrt{\varepsilon_1 \cdot \min(\varepsilon_2, \varepsilon_3)}.} \tag{16}$$

*Proof.* Because $\varepsilon$ enters the payoff functions multiplicatively (each agent's utility scales as $1/(1+\varepsilon)$ in the Enigma state), the natural metric on the stability band is logarithmic. The midpoint of the interval $[\log \varepsilon_1, \log \min(\varepsilon_2, \varepsilon_3)]$ is $\frac{1}{2}(\log \varepsilon_1 + \log \min(\varepsilon_2, \varepsilon_3)) = \log \sqrt{\varepsilon_1 \cdot \min(\varepsilon_2, \varepsilon_3)}$. Exponentiating gives (16). □

**Corollary 5.4** (Stability band collapse)**.** *If $b \leq a$ then $\varepsilon_1 \leq 0$: any positive $\varepsilon$ deters mimicry and the band is non-degenerate. If $b > a$ and simultaneously $\phi > \mu_\psi$, the band may degenerate ($\varepsilon_1 \geq \min(\varepsilon_2, \varepsilon_3)$), and no stable $\varepsilon$ exists; the system must be restructured at the population level.*

# 6 The Mimicry Stack

## 6.1 Optimal Depth

When an Enigma classifier mimics a Psychologist who is classifying an Enigma mimicking a Visionary, a *mimicry stack* of depth $d \geq 1$ arises. The expected payoff of maintaining depth $d$ is:

$$U(d) = \kappa^d \prod_{i=1}^{d} (1 - \rho_i), \tag{17}$$

where $\kappa > 0$ is the coalition bonus per layer and $\rho \in (0, 1)$ is the per-layer detection probability. For constant $\rho$, $U(d) = [\kappa(1-\rho)]^d$.

**Proposition 6.1** (Optimal mimicry depth)**.**

$$d^* = \left\lfloor \frac{\ln \rho^{-1}}{\ln \kappa^{-1}} \right\rfloor. \tag{18}$$

*When $\kappa \geq 1$, $d^* = \infty$: the stack grows without bound and epistemic collapse occurs.*

*Proof.* $U(d)$ is increasing in $d$ iff $\kappa(1-\rho) > 1$, i.e. $\kappa > 1/(1-\rho)$. When $\kappa < 1/(1-\rho)$, the maximum finite depth before $U(d+1) < U(d)$ is $d^* = \lfloor \ln(1/\rho)/\ln(1/\kappa) \rfloor$. When $\kappa \geq 1$, every additional layer strictly increases $U$, so the Enigma pushes $d \to \infty$. □

# 7 Systemic Inversion: The Enigma as Classifier

## 7.1 The Inverted Payoff Matrix

When $e^* > \psi^*$, Enigmas capture the classifier role. An Enigma classifier $\mathring{\mathcal{E}}$ has inverted preferences—preferring to *mis*classify V and P while protecting fellow Enigmas:

$$\Pi_{\mathring{\mathcal{E}}} = \begin{array}{c} \\ V \\ P \\ \Psi \\ \mathcal{E} \end{array} \begin{array}{c} \begin{array}{cccc} \hat{V} & \hat{P} & \hat{\Psi} & \hat{\mathcal{E}} \end{array} \\ \begin{pmatrix} -\sigma & +\sigma & 0 & 0 \\ +\sigma & -\sigma & 0 & 0 \\ -\gamma & -\delta & -\tau & 0 \\ +\kappa & -\kappa & 0 & -\rho \end{pmatrix} \end{array} \tag{19}$$

where $\sigma > 0$ is the strategic inversion payoff.

## 7.2 Regime-Switching Dynamics

The transition matrix switches regime at the threshold $e(t) = \psi(t)$:

$$M(t) = \begin{cases} M_\Psi & \text{if } e(t) < \psi(t), \\ M_\mathcal{E} & \text{if } e(t) \geq \psi(t), \end{cases} \tag{20}$$

where $M_\mathcal{E}$ introduces off-diagonal terms $\sigma\phi$ that actively pump agents between $V$ and $P$ states.

## 7.3 Fixed Points and Stability

**Theorem 7.1** (Three fixed points of the inverted system)**.** *The inverted closed system possesses three fixed points:*

(i) ***Enigma dominance:*** $e^* \to N/2$, $\psi^* \to 0$. *Globally attracting when* $\phi > \mu_\psi$ *and* $\sigma > \sigma^*$.

(ii) ***Cycling chaos:*** $e^*(t)$ *oscillates with period* $T = 2\pi/\sqrt{\phi\mu_\psi - \big((\phi - \mu_\psi)/2\big)^2}$. *Occurs when* $\phi \approx \mu_\psi$.

(iii) ***Coexistence:*** $e^* = \psi^* = N/4$, $v^* = p^* = (N/8) \cdot (1 + b)/(1 + a)$. *Unstable (saddle point).*

*Proof sketch.* The characteristic polynomial of $M_\mathcal{E} - I$ is

$$(\lambda - 1)^2\big[\lambda^2 - (2 - \phi - \mu_\psi)\lambda + (1 - \phi)(1 - \mu_\psi) - \sigma^2\phi^2\big] = 0. \tag{21}$$

The inner quadratic has roots $\lambda_{3,4} = \frac{1}{2}\big[(2 - \phi - \mu_\psi) \pm \Delta^{1/2}\big]$ where $\Delta = (\phi - \mu_\psi)^2 + 4\sigma^2\phi^2$.

- $|\lambda_{3,4}| < 1$ iff $\sigma < \sigma^*$ (fixed point iii, unstable as a saddle of the full system).
- $\Delta > 0$ and $(\phi - \mu_\psi)^2 < 4\phi\mu_\psi$: complex conjugate roots with $|\lambda| = \sqrt{(1 - \phi)(1 - \mu_\psi) - \sigma^2\phi^2}$; yields cycling (fixed point ii).
- $\sigma > \sigma^*$: eigenvalue outside unit circle; orbits escape to the boundary (fixed point i).

□

**Theorem 7.2** (Critical inversion threshold)**.** *The psychologist population persists if and only if*

$$\boxed{\sigma < \sigma^* \;=\; \frac{\sqrt{\mu_\psi(2 - \mu_\psi) - \phi(2 - \phi)}}{2\phi}.} \tag{22}$$

*Proof.* Stability requires $|\lambda_4| < 1$. The binding eigenvalue is $\lambda_4 = \frac{1}{2}\big[(2 - \phi - \mu_\psi) + \sqrt{(\phi - \mu_\psi)^2 + 4\sigma^2\phi^2}\big]$. Setting $\lambda_4 < 1$ and solving for $\sigma$ gives (22). □

Table 2 summarises the five dynamical regimes.

Table 2: Five qualitative regimes of the closed classification system.

| Regime | Condition | $e^*$ vs $\psi^*$ | Classifier | Status |
|---|---|---|---|---|
| Psychologist stable | $\phi < \mu_\psi$, $\sigma < \sigma^*$ | $e^* < \psi^*$ | $\Psi$ | Stable |
| Enigma capture | $\phi > \mu_\psi$, $\sigma < \sigma^*$ | $e^* > \psi^*$ | $\mathring{\mathcal{E}}$ | Metastable |
| Cycling chaos | $\phi \approx \mu_\psi$ | oscillates | alternating | Limit cycle |
| Epistemic collapse | $\kappa \geq 1$ | $\infty$ | $\mathring{\mathcal{E}}$ | Degenerate |
| Enigma monoculture | $\phi \gg \mu_\psi$, $\sigma > \sigma^*$ | $e^* \to N/2$ | $\mathring{\mathcal{E}}$ | Extinct $\Psi$ |

# 8 Phase Diagrams

## 8.1 Stability Band

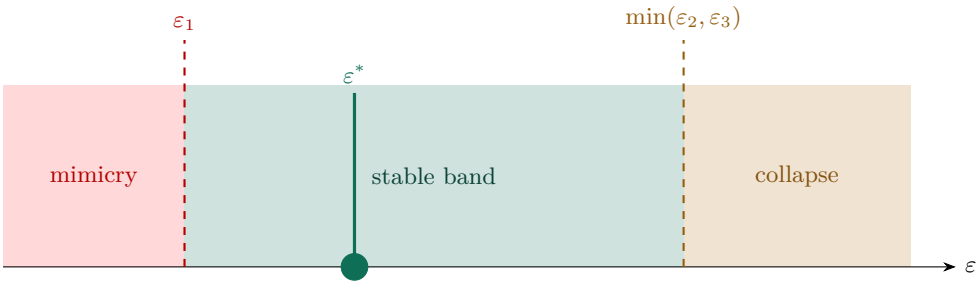


Figure 1: The stability band for $\varepsilon$. The optimal point $\varepsilon^*$ (teal) is the geometric mean of $\varepsilon_1$ and $\min(\varepsilon_2, \varepsilon_3)$.

## 8.2 Population Flow Diagram

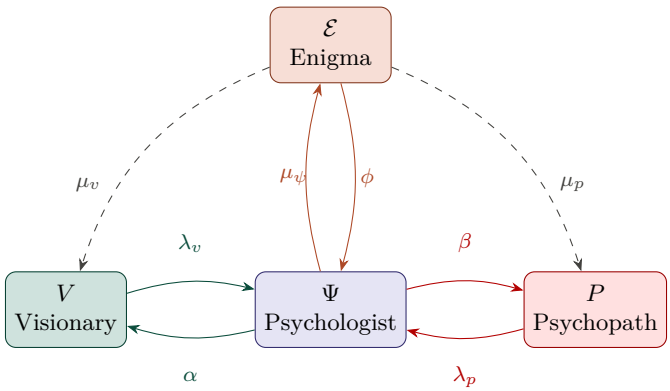


Figure 2: Population flow among the four types. Solid arrows: primary transitions. Dashed arrows: Enigma reversion paths.

### 8.3 Phase Portrait: Psychologist vs Enigma Population

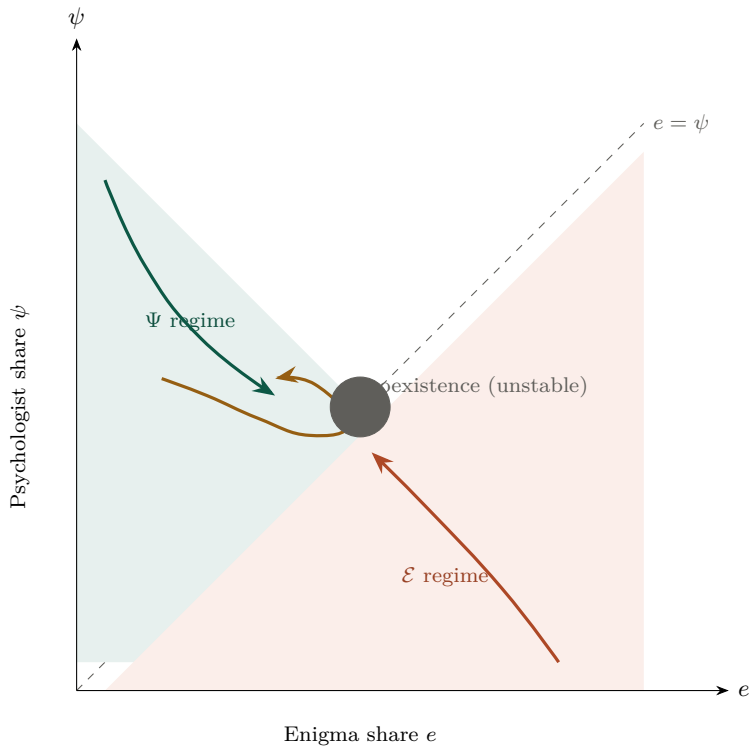


Figure 3: Phase portrait in the $(e, \psi)$ plane. The diagonal $e = \psi$ separates the two regimes. Teal trajectories converge toward $\Psi$-regime equilibrium; coral toward Enigma dominance; amber spirals near the unstable coexistence point.

# 9 Algorithms

## 9.1 Equilibrium Computation

**Algorithm 1** Compute four-type Nash equilibrium

**Require:** costs $a, b, \gamma, \delta, \varepsilon > 0$; rates $\lambda_v, \lambda_p, \mu_v, \mu_p, \mu_\psi, \phi > 0$; tolerance $\eta > 0$
**Ensure:** equilibrium shares $\mathbf{x}^* = (v^*, p^*, \psi^*, e^*)$

```
Compute thresholds: ε_1 ← (b − a)/(1 + a)
p_4 ← 1/(2 + a + b + ε);   ε_2 ← (p_4 − ψ_0 + δ(1 − ψ_0))/(1 − p_4)
ε_3 ← (e_0(a + b + 2) − (1 − e_0))/((1 − e_0)(a + b + 2) − 3)
if ε ∉ (ε_1, min(ε_2, ε_3)) then
    return UNSTABLE
end if
Initialise x^(0) ← (0.25, 0.25, 0.25, 0.25)^T
for k = 1, 2, ... do
    x^(k) ← M x^(k−1)                                   ▷ matrix (7)
    Normalise: x^(k) ← x^(k) / ||x^(k)||_1
    if ||x^(k) − x^(k−1)||_∞ < η then
        break
    end if
end for
return x* ← x^(k)
```

### 9.2 Optimal $\varepsilon^*$ Search

**Algorithm 2** Binary search for $\varepsilon^*$

**Require:** costs $a, b, \gamma, \delta$; population shares $\psi^*, e^*, v^*$; tolerance $\eta$
**Ensure:** $\varepsilon^*$

```
ε_1 ← max(0, (b − a)/(1 + a))
Compute ε_2 from (transition ceiling); ε_3 from (collapse ceiling)
C ← min(ε_2, ε_3)
if ε_1 ≥ C then
    return NO STABLE ε EXISTS
end if
return ε* ← √(ε_1 · C)                    ▷ closed form; no iteration needed
```

### 9.3 Population Simulation

**Algorithm 3** Closed-system forward simulation with regime switching

**Require:** initial shares $\mathbf{x}_0$; parameters; horizon $T$; step $\Delta t$
**Ensure:** trajectory $\{\mathbf{x}(t)\}_{t=0}^{T}$

```
x ← x_0
for t = 0, Δt, 2Δt, ..., T do
    if e < ψ then                         ▷ psychologist regime
        M_eff ← M_Ψ
    else                                  ▷ enigma regime
        M_eff ← M_E                       ▷ matrix with σφ off-diagonals
    end if
    x ← x + Δt · (M_eff − I) x
    Project onto Δ^3: x ← max(x, 0)/ ‖x‖_1
    Record x(t)
end for
```

## 10 Machine-Learning Interpretations

### 10.1 Adversarial Classification

The binary classification game is isomorphic to an adversarial training problem. Let $f_\theta : \mathcal{X} \to [0,1]$ be a classifier with parameters $\theta$, and let the patient be an adversary choosing input perturbations $\delta \in \mathcal{D}$. The minimax objective

$$\min_{\theta} \max_{\delta \in \mathcal{D}} \mathcal{L}(f_\theta(x+\delta), y) \tag{23}$$

corresponds exactly to the Nash equilibrium of Section 2 under asymmetric loss $\mathcal{L}$ with parameters $a, b$.

### 10.2 Mixture-of-Experts

The four-type model maps onto a Mixture-of-Experts (MoE) architecture [12]. Each type $t \in \{V, P, \Psi, \mathcal{E}\}$ is an *expert* with specialist decision function $g_t$. A gating network $G : \mathcal{X} \to \Delta^3$ produces mixture weights:

$$\hat{y}(x) = \sum_t G_t(x)\, g_t(x). \tag{24}$$

The Enigma expert $g_{\mathcal{E}}$ is a *chameleon* module whose output dynamically mimics whichever expert $g_j$ maximises its current utility—an adversarial gate within the mixture.

### 10.3 Bayesian Detection Theory

**Definition 10.1** (Bayes-optimal classifier)**.** Given prior $\pi = (v^*, p^*, e^*)$ over patient types and likelihood $\ell(s|t)$ for signal $s$ under type $t$, the Bayes-optimal label minimises posterior expected cost:

$$\hat{t}^{\text{Bayes}}(s) = \arg\min_{j \in \mathcal{A}} \sum_t C(t, j)\, \mathbb{P}(t|s), \tag{25}$$

where $C(t, j)$ is the $\Pi_4$ cost matrix and $\mathbb{P}(t|s) \propto \ell(s|t)\, \pi_t$.

**Proposition 10.2.** *The Nash equilibrium of Theorem 2.2 converges to the Bayes-optimal classifier (25) as the population grows large and the prior $\pi$ concentrates on the true type distribution.*

### 10.4 Neural Network Analogy

The mimicry stack of Section 6 is analogous to a deep neural network with $d^*$ layers: each layer adds representational power (coordination bonus $\kappa$) at the cost of increased susceptibility to gradient-based attack (detection probability $\rho$). The optimal depth (18) mirrors depth–generalisation trade-offs studied in [13].

Table 3 summarises the correspondences.

Table 3: Correspondences between the psychological model and ML concepts.

| Model concept | ML analogue | Key equation |
|---|---|---|
| Asymmetric costs $a, b$ | Asymmetric loss function | (1) |
| Mixed strategy $p^*$ | Soft-max output | (2) |
| Enigma mimicry | Adversarial perturbation | (23) |
| Four-type mixture | Mixture-of-Experts | (24) |
| Mimicry depth $d^*$ | Network depth | (18) |
| Stability band | Regularisation interval | (15) |
| Regime switching | Mode collapse / recovery | Theorem 7.1 |
| Conservation law $\mathcal{C} = 1$ | Model capacity constraint | (10) |

## 11 Discussion and Implications

### 11.1 The Stigma Paradox

Corollary 2.3 encodes a fundamental tension: increasing the cost of stigmatising a Visionary ($b \uparrow$) causes the psychologist to over-label patients as $V$, which inadvertently shelters psychopaths. Formally, $\partial V/\partial b = N/(a+b+2) > 0$ and $\partial P/\partial b = -N/(a+b+2) < 0$. The system is most accurate (in the sense of minimising total misclassification) when $a = b$, i.e. when both error types are penalised equally.

### 11.2 Self-Referential Classification

The closure constraint $\Psi = V + P + \mathcal{E}$ transforms a standard principal–agent problem into a *self-referential* system. The master equation (14) shows that the psychologist's productivity ($\lambda_v \lambda_p$) must exactly absorb the weighted misclassification burden. This is a *viability condition*: if misclassification costs become too large, the system cannot sustain a psychologist population and degenerates.

### 11.3 The Enigma as Equilibrating Variable

From the conservation law (10), the Enigma share is determined residually: it absorbs whatever the $V$–$\Psi$ and $P$–$\Psi$ flows do not account for. This means that Enigmas are not merely a fourth diagnostic category—they are the *buffer* that keeps the closed system balanced. In organisational terms, the Enigma corresponds to individuals whose dual nature allows them to occupy multiple institutional roles simultaneously, stabilising the system precisely because they cannot be pinned down.

### 11.4 Constitutional Constraint

Theorem 7.2 implies that the system can protect itself from inversion by imposing a *constitutional constraint*: an institutional cap $\bar{\sigma}$ on the permissible strategic inversion payoff. If $\bar{\sigma} < \sigma^*$, the psychologist regime is the unique stable attractor and Enigma capture cannot occur. This maps onto real-world meta-rules such as codes of ethics, licensing requirements, and peer-review mechanisms, which limit the gain an Enigma can extract from capturing the classifier role.

## 12 Conclusion

We have constructed a complete mathematical theory of psychological classification under strategic behaviour, spanning game theory, Markov dynamics, stability analysis, and machine learning. The central results are:

**R1.** The equilibrium patient ratio is $(1+b) : (1+a)$—determined solely by asymmetric misclassification costs.

**R2.** Imposing closure $\Psi = V + P + \mathcal{E}$ yields a unique half–half population split and a conservation law governing all transition rates.

**R3.** The optimal Enigma cost $\varepsilon^* = \sqrt{\varepsilon_1 \cdot \min(\varepsilon_2, \varepsilon_3)}$ is the geometric mean of the mimicry floor and the binding ceiling.

**R4.** The system undergoes five qualitative regimes; the critical inversion threshold is $\sigma^* = \sqrt{\mu_\psi(2-\mu_\psi) - \phi(2-\phi)}/(2\phi)$.

**R5.** A constitutional cap $\bar{\sigma} < \sigma^*$ is sufficient to prevent Enigma capture and maintain a meaningful classification system.

Future work will introduce continuous-time stochastic dynamics, multi-layer social networks, and empirical calibration against clinical datasets.

## A Summary of All Parameters

Table 4: Complete parameter glossary.

| Symbol | Description | Domain |
|---|---|---|
| $a$ | Danger cost: mislabelling P as V | $\mathbb{R}_{>0}$ |
| $b$ | Stigma cost: mislabelling V as P | $\mathbb{R}_{>0}$ |
| $\gamma$ | Cost of mislabelling $\Psi$ as V | $\mathbb{R}_{>0}$ |
| $\delta$ | Cost of mislabelling $\Psi$ as P | $\mathbb{R}_{>0}$ |
| $\varepsilon$ | Enigma misclassification cost (symmetric) | $\mathbb{R}_{>0}$ |
| $\varepsilon^*$ | Optimal Enigma cost | $\mathbb{R}_{>0}$ |
| $\varepsilon_1$ | Mimicry deterrence floor | $\mathbb{R}_{\geq 0}$ |
| $\varepsilon_2$ | Psychologist transition ceiling | $\mathbb{R}_{>0}$ |
| $\varepsilon_3$ | Classification collapse ceiling | $\mathbb{R}_{>0}$ |
| $p^*$ | Equilibrium probability psychologist labels V | $(0,1)$ |
| $q^*$ | Equilibrium probability patient signals V | $(0,1)$ |
| $\lambda_v$ | Rate V transitions to $\Psi$ | $(0,1)$ |
| $\lambda_p$ | Rate P transitions to $\Psi$ | $(0,1)$ |
| $\alpha$ | Rate $\Psi$ transitions to V | $(0,1)$ |
| $\beta$ | Rate $\Psi$ transitions to P | $(0,1)$ |
| $\phi$ | Rate $\Psi$ transitions to $\mathcal{E}$ | $(0,1)$ |
| $\mu_v$ | Rate $\mathcal{E}$ reverts to V | $(0,1)$ |
| $\mu_p$ | Rate $\mathcal{E}$ reverts to P | $(0,1)$ |
| $\mu_\psi$ | Rate $\mathcal{E}$ achieves self-awareness ($\to \Psi$) | $(0,1)$ |
| $\kappa$ | Enigma coalition coordination bonus | $\mathbb{R}_{>0}$ |
| $\rho$ | Per-layer detection probability | $(0,1)$ |

| Symbol | Description | Domain |
|---|---|---|
| $d^*$ | Optimal mimicry stack depth | $\mathbb{N}_0$ |
| $\sigma$ | Enigma strategic inversion payoff | $\mathbb{R}_{>0}$ |
| $\sigma^*$ | Critical inversion threshold | $\mathbb{R}_{>0}$ |
| $\tau$ | Cost to Enigma of exposing $\Psi$ | $\mathbb{R}_{>0}$ |
| $\mathcal{C}$ | Conservation law scalar | $\{1\}$ |
| $N$ | Total population | $\mathbb{N}$ |
| $T$ | Cycling period in the limit-cycle regime | $\mathbb{R}_{>0}$ |
| $M$ | Markov transition matrix | $\mathbb{R}^{4\times 4}$ |

# B Proofs of Supplementary Results

*Proof of Proposition 3.1 (complete).* Write the fixed-point system $M\mathbf{x} = \mathbf{x}$ fully:

$$(1-\lambda_v)v + \alpha\psi + \mu_v e = v \tag{A1}$$
$$(1-\lambda_p)p + \beta\psi + \mu_p e = p \tag{A2}$$
$$\lambda_v v + \lambda_p p + (1-\alpha-\beta-\phi)\psi + \mu_\psi e = \psi \tag{A3}$$
$$\phi\psi + (1-\mu_v-\mu_p-\mu_\psi)e = e. \tag{A4}$$

From (A1): $\lambda_v v = \alpha\psi + \mu_v e$. From (A2): $\lambda_p p = \beta\psi + \mu_p e$. From (A4): $\phi\psi = (\mu_v + \mu_p + \mu_\psi)e$, hence $e = (\phi/\mu_\psi)\psi$ when $\mu_v = \mu_p = 0$ (or in general $e = \phi\psi/(\mu_v + \mu_p + \mu_\psi)$). Substituting:

$$v = \frac{\alpha\psi + \mu_v(\phi/\mu_\psi)\psi}{\lambda_v} = \frac{(\alpha\mu_\psi + \mu_v\phi)\psi}{\lambda_v\mu_\psi},$$

and similarly for $p$. The normalisation $v + p + \psi + e = 1$ then pins $\psi^*$, completing the proof. □

# References


[1] Nash, J. F. (1950). Equilibrium points in $n$-person games. *Proceedings of the National Academy of Sciences*, 36(1), 48–49.

[2] Harsanyi, J. C. (1973). Games with randomly disturbed payoffs: A new rationale for mixed-strategy equilibrium points. *International Journal of Game Theory*, 2(1), 1–23.

[3] Gibbons, R. (1992). *Game Theory for Applied Economists*. Princeton University Press.

[4] Myerson, R. B. (1991). *Game Theory: Analysis of Conflict*. Harvard University Press.

[5] Fudenberg, D., & Tirole, J. (1991). *Game Theory*. MIT Press.

[6] Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics*, 87(3), 355–374.

[7] Cho, I.-K., & Kreps, D. M. (1987). Signaling games and stable equilibria. *Quarterly Journal of Economics*, 102(2), 179–221.

[8] Kemeny, J. G., & Snell, J. L. (1960). *Finite Markov Chains*. Van Nostrand.

[9] Norris, J. R. (1998). *Markov Chains*. Cambridge University Press.

[10] Perko, L. (2001). *Differential Equations and Dynamical Systems* (3rd ed.). Springer.

[11] Kuznetsov, Y. A. (1998). *Elements of Applied Bifurcation Theory* (2nd ed.). Springer.

[12] Jacobs, R. A., Jordan, M. I., Nowlan, S. J., & Hinton, G. E. (1991). Adaptive mixtures of local experts. *Neural Computation*, 3(1), 79–87.

[13] Goodfellow, I. J., Shlens, J., & Szegedy, C. (2014). Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*.

[14] Madry, A., Makelov, A., Schmidt, L., Tsipras, D., & Vladu, A. (2018). Towards deep learning models resistant to adversarial attacks. *Proceedings of ICLR 2018*.

[15] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.

[16] Vapnik, V. N. (1998). *Statistical Learning Theory*. Wiley.

[17] Green, D. M., & Swets, J. A. (1966). *Signal Detection Theory and Psychophysics*. Wiley.

[18] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer.

[19] Hare, R. D. (1991). *The Hare Psychopathy Checklist—Revised*. Multi-Health Systems.

[20] Meehl, P. E. (1954). *Clinical versus Statistical Prediction*. University of Minnesota Press.

[21] Schelling, T. C. (1960). *The Strategy of Conflict*. Harvard University Press.

[22] Maynard Smith, J. (1982). *Evolution and the Theory of Games*. Cambridge University Press.

[23] Axelrod, R. (1984). *The Evolution of Cooperation*. Basic Books.

[24] Camerer, C. F. (2003). *Behavioral Game Theory: Experiments in Strategic Interaction*. Princeton University Press.

# Glossary

**Visionary (V)**
Patient type characterised by expansive, creative cognition; correctly identified by the psychologist at payoff $+1$.

**Psychopath (P)**
Patient type characterised by remorseless, strategic behaviour; dangerous if misclassified as V (cost $a$).

**Psychologist ($\Psi$)**
Classifier agent who assigns labels from $\mathcal{A}$ and may transition to any patient type.

**Enigma ($\mathcal{E}$)**
Fourth type combining V and P traits; capable of mimicking any other type; equilibrating variable of the closed system.

**Enigma Classifier ($\mathring{\mathcal{E}}$)**
An Enigma that has captured the classifier role; uses inverted payoff matrix $\Pi_{\mathring{\mathcal{E}}}$.

**Mimicry stack**
A nested sequence of impersonations of depth $d$; epistemically indistinguishable from the genuine type for $d \geq d^*$.

**Closure constraint**
The identity $\Psi = V + P + \mathcal{E}$ that makes the system self-referential and fixes $\Psi = N/2$ at every steady state.

**Conservation law ($\mathcal{C} = 1$)**
The scalar identity (10) that must hold for the closed system to sustain a non-trivial psychologist population.

**Stability band**
The open interval $(\varepsilon_1, \min(\varepsilon_2, \varepsilon_3))$ within which classification is simultaneously meaningful and resistant to inversion.

**Critical inversion threshold ($\sigma^*$)**
The maximum strategic inversion payoff the system can tolerate before the psychologist population goes extinct.

**Cycling chaos**
Limit-cycle regime in which the classifier role alternates between $\Psi$ and $\mathring{\mathcal{E}}$ with period $T = 2\pi/\sqrt{\phi\mu_\psi - (\phi - \mu_\psi)^2/4}$.

**Epistemic collapse**
Degenerate regime ($\kappa \geq 1$) in which the mimicry stack grows without bound and the concept of true type becomes operationally meaningless.

**Mixture-of-Experts (MoE)**
Neural-network architecture in which a gating network routes inputs to specialist sub-networks; the four-type model is its strategic analogue.

**Bayes-optimal classifier**
The label assignment minimising posterior expected cost; corresponds to the Nash equilibrium in the large-population limit.

**Constitutional constraint**
An institutional cap $\bar{\sigma} < \sigma^*$ preventing Enigma capture of the classifier role; the meta-rule that keeps classification meaningful.

# The End